SMARTBRIDGE
Let's Bridge the Gap

SKILL WALLET
A SmartBridge Product

Project Initialization and Planning Phase

Date	2 July 2026
Team ID	XXXXX
Project Title	OPTICROP – SMART AGRICULTURAL PRODUCTION OPTIMIZATION ENGINE
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal report aims to transform loan approval using machine learning, boosting efficiency and accuracy. It tackles system inefficiencies, promising better operations, reduced risks, and happier customers. Key features include a machine learning-based credit model and real-time decision-making.

Project Overview	
Objective	The primary objective of this project is to assist farmers in selecting the most suitable crop based on soil nutrients and environmental conditions using Machine Learning techniques. The system aims to improve agricultural productivity, reduce crop failure risks, and support data-driven farming decisions.
Scope	The project focuses on analyzing soil and weather parameters such as Nitrogen (N), Phosphorous (P), Potassium (K), Temperature, Humidity, pH, and Rainfall to recommend the most appropriate crop. The solution includes machine learning model development, prediction generation, and a user-friendly web application interface.
Problem Statement	
Description	Farmers often face difficulties in selecting the right crop due to variations in soil nutrients and environmental conditions. Traditional crop selection methods depend heavily on experience and may not

	always produce optimal results, leading to reduced productivity and financial losses.
Impact	Providing accurate crop recommendations can improve agricultural productivity, increase profitability, reduce resource wastage, and support sustainable farming practices.
Proposed Solution	
Approach	The proposed solution uses Machine Learning algorithms to analyze agricultural data and predict the most suitable crop based on soil and weather conditions. A trained model processes user-provided inputs and generates crop recommendations through a web-based application.
Key Features	Machine Learning-based crop recommendation system. User-friendly web interface for data input. Real-time crop prediction generation. Analysis of soil nutrient and weather parameters. High prediction accuracy using trained models. Scalable architecture for future enhancements. Support for future weather API integration. Provision for fertilizer recommendation in future versions

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU for model training and prediction	Intel i5/i7 Processor
Memory	RAM specifications	8 GB
Storage	Disk space for dataset, models, and application files	256 GB SSD
Software		

Frameworks	Web Framework	Flask
Libraries	Data Processing & Machine Learning	Scikit-Learn, Pandas, NumPy, Matplotlib, Seaborn, Joblib
Development Environment	IDE / Code Editor	VS Code
Data		
Dataset	Crop Recommendation Dataset	CSV Format
Source	Agricultural Dataset	Kaggle Dataset
Features	Soil & Weather Parameters	N, P, K, Temperature, Humidity, pH, Rainfall
Records	Training Samples	Approximately 2200+ Records